CSE-400 Scribe - L23

L23_Amazone Sale_20...
PDF

Generate a lecture scribe intended strictly as exam-oriented reference material for CSE 400. Use only the provided context (lecture slides/pdfs and the explicitly relevant textbook parts). Treat this context as authoritative and complete. Do not use or introduce outside knowledge.

When processing handwritten notes, read them carefully and conserve their meanings. Interpret symbols, notation, and equations with precision. Resolve ambiguities strictly using consistency with the lecture slides and textbook content. Draw figures/diagrams when needed. Do not guess or hallucinate unclear material.

Write in a formal, academic tone suitable for closed-notes exam revision. Preserve the lecture's structure, ordering, notation, terminology, and scope exactly. While writing the scribe, reason step by step through the material exactly as presented in the context, and include that reasoning explicitly in the scribe.

Emphasize clarity of dependencies between assumptions, definitions, and results without introducing new explanations.

Focus more on: 1-State and explain all definitions, notation, and assumptions explicitly, 2-Present theorems, propositions, and results exactly as given, 3-Present proofs and derivations step by step, preserving all logical dependencies and, 4-Present worked examples step by step, showing all intermediate steps exactly as taught.

Content constraints: 1-Include only material present in the provided context, 2-Do not introduce new examples, interpretations, intuitions, alternative arguments, or additional explanations, 3-Do not omit conditions,

assumptions, or logical steps present in the source material and, 4-Do not simplify, generalize, or paraphrase beyond faithful reconstruction.

Organization Note: Structure the scribe with clear sections and headings reflecting the lecture flow so that a student relying solely on this document can accurately reconstruct definitions, derivations, proofs, and example solutions for exam preparation.

Lecture 23: Predicting the Impossible — Tail Bounds and System Reliability

(Amazon Black Friday Sale 2026 Case Study — Coding Session: Tails and Bounds)

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1. Problem Context: System Reliability Under Uncertainty

1.1 Scenario Definition

- The system models **Amazon Black Friday Sale 2026 traffic**.
- High-volume distributed systems (datacenters) must handle **random request arrivals**.

From *page 3 diagram*:

- Requests arrive at servers as **random processes**.
- Modeled using **Poisson arrivals**:

$$\text{Number of arrivals in time interval} \sim \text{Poisson}(\lambda)$$

- Critical system event:

$$\text{Server overload occurs if } X \geq k$$

  where:

  - X: number of requests in a time window
  - k: system crash threshold

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2. Tail Probability as Failure Metric

2.1 Definition

The system reliability question reduces to computing:

$$P(X \geq k)$$

From *page 4–5 visualization*:

- Mean behavior appears stable.
- Failures occur due to **rare tail events**.
- Key statement:
  - *"The mean is safe. The tail is catastrophic."*

2.2 Example from Simulation (Page 5)

- Poisson traffic: $\lambda = 50$
- Crash threshold: $k = 60$
- Empirical crash rate:

$$8.7\% \quad (87 \text{ crashes in } 1000 \text{ minutes})$$

Thus:

$$\text{Goal: Estimate } P(X \geq 60)$$

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3. Necessity of Probability Bounds

Exact computation of $P(X \geq k)$ may not always be feasible.
Thus, **upper bounds** are used.

From *page 6 hierarchy*:

Level	Method	Required Information
1	Markov	Mean μ
2	Chebyshev	Mean μ, Variance σ^2
3	Chernoff	Full distribution (MGF)

Observation:
More information ⇒ tighter bounds.

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4. Markov's Inequality

4.1 Statement

For non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

4.2 Inputs

- Only requires:

$$\mu = E[X]$$

4.3 Behavior (Page 7)

- Decay:

$$\frac{1}{k} \quad (\text{linear decay})$$

- Strong overestimation:
  - Example shown:
    - Exact: 0.004
    - Markov: 0.714

4.4 Interpretation

- Bound is **loose**.
- Overestimates heavily.
- Wastes system resources due to pessimism.

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5. Chebyshev's Inequality

5.1 Statement

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Equivalent tail form (as shown in slides):

$$P(X \geq a) \leq \frac{\sigma^2}{(a - \mu)^2}$$

5.2 Inputs

- Mean:

$$\mu = E[X]$$

- Variance:

$$\sigma^2 = \mathrm{Var}(X)$$

5.3 Behavior (Page 8)

- Decay:

$$\frac{1}{k^2} \quad \text{(polynomial decay)}$$

5.4 Example

- Exact: 0.004
- Chebyshev: 0.125

5.5 Interpretation

- Tighter than Markov.
- Still overestimates significantly.

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6. Chernoff Bound

6.1 Statement (General Form)

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

where:

$$M_X(t) = E[e^{tX}] \quad \text{(Moment Generating Function)}$$

6.2 Alternate Form (Page 9)

$$P(X \geq a) \leq e^{-t(a-\mu)} \quad \text{(after substitution and optimization)}$$

6.3 Inputs

- Full distribution via MGF.

6.4 Behavior

- Decay:

$$e^{-k} \quad \text{(exponential decay)}$$

6.5 Example

- Exact: 0.092

- Chernoff: 0.391 *(tightest among bounds in slide context)*

6.6 Interpretation

- Tightest bound.
- Closely follows actual tail probability.

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7. Comparative Decay Rates (Page 10)

Method	Decay Type	Expression
Markov	Linear	$1/k$
Chebyshev	Polynomial	$1/k^2$
Chernoff	Exponential	e^{-k}

Conclusion:

$$\text{Chernoff} \ll \text{Chebyshev} \ll \text{Markov}$$

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8. Case Study: SLA Breach (Page 11)

8.1 Parameters

- Poisson load:

$$\lambda = 500$$

- SLA threshold:

$$k = 600$$

8.2 Objective

$$\text{Estimate } P(X \geq 600)$$

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9. Cost of Loose Bounds (Page 12)

Method	Estimated Failure Probability
Exact	1.4%

Method	Estimated Failure Probability
Chernoff	8.8%
Chebyshev	20%
Markov	90.9%

Observation

- Markov exaggerates risk drastically.
- Leads to incorrect system design decisions.

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10. Financial Interpretation (Pages 13–14)

10.1 Waste Multiplier Definition

$$\text{Waste Multiplier} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

10.2 Example at $\lambda = 500$

- Chernoff: $\approx 10.7\times$ waste
- Chebyshev: $\approx 6{,}422\times$ waste
- Markov: $\approx 107{,}039\times$ waste

Conclusion

$$\text{Loose bound} \Rightarrow \text{Over-provisioning} \Rightarrow \text{Financial loss}$$

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11. Distribution Sensitivity (Page 15)

Observation

- Chernoff performs well for:
  - Poisson
  - Binomial
- For heavy-tailed distributions (e.g., Gamma):
  - Chernoff may overestimate.
  - Alternative methods or simulation required.

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12. Engineer's Decision Matrix (Page 16)

Method	Requires	Decay	Tightness	Use Case
Markov	Mean μ	$1/k$	Loose	Only mean known
Chebyshev	μ, σ^2	$1/k^2$	Moderate	Variance known
Chernoff	Full MGF	e^{-k}	Tight	Precise SLA sizing

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13. Class Exercise: Laplace Distribution (Pages 17–18)

13.1 Given

$$X \sim \text{Laplace}(\mu, b)$$

- Standard deviation:

$$\sigma = b\sqrt{2}$$

13.2 Objective

Compute Chernoff bound for:

$$a = \mu + 3\sigma$$

13.3 Derivation Blueprint

Step 1: Compute MGF

$$M_X(t) = E[e^{tX}]$$

Step 2: Apply Chernoff

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimize

- Differentiate w.r.t t
- Set derivative = 0
- Solve for optimal t^*

Step 4: Substitute

$$P(X \geq a) \leq e^{-t^* a} M_X(t^*)$$

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14. Final Synthesis

1. System failures are governed by:

$$P(X \geq k)$$

2. Exact computation often infeasible ⇒ bounds required.
3. Trade-off:
   - Less information ⇒ weaker bound ⇒ higher cost.
   - More information ⇒ tighter bound ⇒ efficient system design.
4. Mathematical decay governs engineering decisions:

$$\text{Linear} \to \text{Polynomial} \to \text{Exponential}$$

5. Core principle:

$$\text{Better probabilistic modeling} \Rightarrow \text{Lower infrastructure cost}$$